UNIT 1 · TOPIC 6(C)

Mowing Equipment

Syllabus topic: Equipment for harvesting, threshing, mowing, chaff cutting and baling

Cutter-bar (reciprocating) mower (Kepner ch. 14)

Parts: cutter bar; knife (sections riveted to the knife back); guards (fingers) with ledger plates; wearing plates; knife clips; grass board & stick; shoes (inner/outer); pitman; flywheel.

Key dimensions and adjustments

- **Pitch** of knife sections and guards = **76.2 mm (3 in.)**; knife stroke = **76.2 mm** (one section pitch).
- **Knife velocity:** crank speed **800–1000 rpm**; average knife speed ≈ **2 × stroke × rpm**.
- **Registration:** the knife section centre coincides with the guard centre at each end of the stroke. Adjust it by shifting the cutter bar in or out.
- **Alignment (lead):** give the cutter bar a forward lead of **~2% of bar length (≈2 cm per metre)** to offset its backward deflection while cutting.
- **Cutting action:** shear between the section edge and the ledger plate.

Other mowers

Rotary and flail mowers cut by **impact**; the forage flail harvester works the same way.

Practice MCQs

Q46. The pitch of knife sections and guards in a conventional mower cutter bar is:

(a) 50.8 mm (b) 76.2 mm
(c) 101.6 mm (d) 88.9 mm

Q47. "Registration" of a mower knife means:

(a) Knife lead adjustment (b) Section centre coinciding with guard centre at stroke ends
(c) Knife tension (d) Serial number stamping

Q48. Lead given to the mower cutter bar is about:

(a) 2% of bar length (b) 10% of bar length
(c) 25 mm regardless of length (d) None

Q49. Cutting in a cutter-bar mower occurs by:

(a) Impact (b) Shear between section and ledger plate
(c) Tearing (d) Bending

Answers

Q46 — (b) 76.2 mm — *Standard cutter-bar geometry — 76.2 mm pitch.*

Q47 — (b) Section centre coinciding with guard centre at stroke ends — *Registration and alignment (lead) are different adjustments.*

Q48 — (a) 2% of bar length — *Registration and alignment (lead) are different adjustments; lead ≈ 2% of bar length.*

Q49 — (b) Shear between section and ledger plate — *Cutter-bar mowers cut by shear.*